Abhinav Vummidichetty

+1 (669) 649-1676 | vummidichettyabhinav@gmail.com | LinkedIn | GitHub | Portfolio

TECHNICAL SKILLS

Programming Languages: Python, **SQL**, Bash, JavaScript/TypeScript
Frameworks & Libraries: PyTorch, **LangGraph**, FastAPI, OpenCV, HuggingFace, **sqlglot**, faster-whisper, Pandas, React
Infrastructure & Databases: Docker, **Qdrant**, MongoDB, PostgreSQL, SQLite, Cloudflare R2, Git, Linux, Apache Airflow
Applied Methodologies: Multimodal RAG, **Autonomous AI Agents**, Computer Vision, Anomaly Detection, AST Parsing, Model Evaluation

EDUCATION

San Jose State University | *M.S. in Data Analytics* Dec 2026
Coursework: Machine Learning, Statistical Modeling, Deep Learning, Generative AI

Andhra University College of Engineering | *B.Tech in Computer Science* May 2024

EXPERIENCE

Extern (Pfizer project) Sep 2026 – Oct 2026
AI-Powered Document Insights & Data Extraction Externship

- Built a **Gradio** RAG assistant with PyMuPDF, Tesseract, BGE, FAISS, and Mistral to search pharmaceutical documents and generate answers with supporting source passages.
- Compared 3 OCR engines, 3 embedding models, and 3 RAG configurations; selected Tesseract after checking identifiers, units, and text grouping against a 300-DPI certificate.
- Reduced the sample workflow's worst-case API calls by **94.7%** with batched classification and boundary detection; cached validated metadata for reuse.

PROJECTS

SeaQL — Text-to-SQL Agent [Demo] *LangGraph, FastAPI, SQLite*

- Architected an autonomous text-to-SQL agent with **LangGraph** and **FastAPI**, implementing **sqlglot** AST parsing, schema injection, and self-healing error recovery across relational tables.
- Achieved **73.1%** query execution accuracy on Olist e-commerce benchmarks with **1.22 s** median latency, securing database integrity by gating 100% of destructive operations with approval tokens.

MedBot — Appointment Workflow Agent [Demo] *LangGraph, MCP, FastAPI*

- Developed a multi-agent clinical reception orchestrator with **LangGraph** and **MCP**, executing autonomous tool-calling routines to query doctor schedules and persist ACID bookings in **MongoDB**.
- Achieved **97.5%** intent routing accuracy and **100%** multi-turn booking completion, integrating deterministic policy verification to block medical advice and route zero-latency emergency triage.

CiteCanvas — Document RAG Workspace [Demo] *Python, Qdrant, VLMs, R2*

- Shipped a production multimodal document workspace using **NVIDIA Vision-Language APIs**, **Cloudflare R2**, and **Qdrant**, extracting coordinate bounding boxes to eliminate hallucinations.
- Achieved **0.073 s** preview response latency with **91.6%** rendering efficiency via tiered 180-DPI rasterization, retaining on-demand lossless 300-DPI rendering for evidence verification.

MomentFlow — Multimodal Video Search [Demo] *SigLIP 2, FastAPI, OCR*

- Built an asynchronous multimodal video discovery engine fusing **SigLIP 2** ZeroGPU embeddings, **faster-whisper** transcriptions, and **Tesseract OCR** into a unified SQLite FTS5 search index.
- Achieved **90.0% Recall@1** and **95.0% Recall@5** on the MSR-VTT benchmark with **59 ms** warm query latency, clustering continuous streams into 5-second semantic chapters with frame seeking.

Argus — Visual Anomaly Detection [GitHub] *PyTorch, ResNet-18, OpenCV*

- Engineered an unsupervised visual inspection edge pipeline with **PyTorch** and **ResNet-18** PatchCore, extracting mid-level patch features and compressing memory banks via greedy coreset subsampling.
- Attained **0.9922 image AUROC** and **95.7%** defect detection accuracy on held-out MVTec industrial components, isolating micro-defects through nearest-neighbor heatmaps and aspect-preserving letterboxing.